

Lab 1: Understanding ORM with a Retail Inventory System

Aim:

To understand Object Relational Mapping (ORM), compare Entity Framework Core with Entity Framework, learn the latest features of EF Core 8.0, and create a .NET Console Application with the required EF Core packages.

1. What is ORM?

ORM (Object Relational Mapping) is a programming technique that maps objects in C# to tables in a relational database.

Instead of writing SQL queries manually, developers work with C# classes and objects. The ORM framework automatically converts object operations into SQL queries.

Example:

C# Class

```
public class Product
{
    public int ProductId { get; set; }
    public string ProductName { get; set; }
    public decimal Price { get; set; }
}
```

SQL Table

Products

```
-----
ProductId    INT
ProductName  VARCHAR(100)
Price        DECIMAL(10,2)
```

Entity Framework Core maps the **Product** class to the **Products** table automatically.

Benefits of ORM

1. Productivity

- Less code to write.
- No need for manual SQL queries.
- Faster development.

2. Maintainability

- Easy to modify classes.
- Database changes are managed using migrations.
- Cleaner code structure.

3. Abstraction from SQL

- Developers work with C# objects.
- SQL statements are generated automatically.
- Reduces database-specific coding.

2. EF Core vs EF Framework

Feature	EF Core	EF Framework (EF6)
Platform	Cross-platform	Windows only
Performance	Faster	Slower
Open Source	Yes	Yes
LINQ Support	Yes	Yes
Async Queries	Supported	Limited
Compiled Queries	Yes	No
JSON Support	Yes	No
Cloud Ready	Yes	Limited
Latest Features	Frequently Updated	Maintenance Mode

EF Core

- Lightweight
- Cross-platform
- High performance
- Supports Linux, macOS, Windows
- Modern architecture

EF Framework (EF6)

- Windows only
- Mature framework
- Mainly used for older .NET Framework applications

3. EF Core 8.0 Features

1. JSON Column Mapping

Allows storing complex objects directly inside SQL Server JSON columns.

Example:

```
public class Product
{
    public int ProductId { get; set; }

    public ProductDetails Details { get; set; }
}
```

Where **Details** can be stored as JSON.

2. Improved Performance

- Compiled Models
- Faster startup
- Reduced memory usage
- Better query execution

3. Interceptors

Interceptors allow executing custom logic before or after database operations.

Example:

- Logging
- Security
- Performance monitoring

4. Better Bulk Operations

EF Core 8 performs:

- Faster Insert
- Faster Update
- Faster Delete

with optimized SQL generation.

4. Create a .NET Console Application

```
PS C:\Users\suman\OneDrive\Desktop> cd ..
PS C:\Users\suman\OneDrive\Desktop> dotnet new console -n RetailInventory
The template "Console App" was created successfully.

Processing post-creation actions...
Restoring C:\Users\suman\OneDrive\Desktop\RetailInventory\RetailInventory.csproj:
Restore succeeded.
```

5. Install EF core Package.

```
PS C:\Users\suman\OneDrive\Desktop\RetailInventory> dotnet add package Microsoft.EntityFrameworkCore.SqlServer

Build succeeded in 1.6s
info : X.509 certificate chain validation will use the default trust store selected by .NET for code signing.
info : X.509 certificate chain validation will use the default trust store selected by .NET for timestamping.
info : Adding PackageReference for package 'Microsoft.EntityFrameworkCore.SqlServer' into project 'C:\Users\suman\OneDrive\Desktop\RetailInventory\RetailInventory.csproj'.
info : CACHE https://api.nuget.org/v3/registration5-semver2/microsoft.entityframeworkcore.sqlserver/index.json
info : CACHE https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.sqlserver/page/0.0.1-alpha/3.1.2.json
info : CACHE https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.sqlserver/page/3.1.3/6.0.0-preview.6.21352.1.json
info : CACHE https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.sqlserver/page/6.0.0-preview.7.21378.4/7.0.17.json
info : CACHE https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.sqlserver/page/7.0.18/9.0.13.json
info : CACHE https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.sqlserver/page/9.0.14/11.0.0-preview.6.26359.118.json
info : Restoring packages for C:\Users\suman\OneDrive\Desktop\RetailInventory\RetailInventory.csproj...
info : CACHE https://api.nuget.org/v3/vulnerabilities/index.json
info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.07.24.05.37.22/vulnerability.base.json
info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.07.24.05.37.22/2026.07.24.05.37.22/vulnerability.update.json
error: NU1202: Package Microsoft.EntityFrameworkCore.SqlServer 10.0.10 is not compatible with net9.0 (.NETCoreApp,Version=v9.0). Package Microsoft.EntityFrameworkCore.SqlServer 10.0.10 supports: net10.0 (.NETCoreApp,Version=v10.0)
error: Package 'Microsoft.EntityFrameworkCore.SqlServer' is incompatible with 'all' frameworks in project 'C:\Users\suman\OneDrive\Desktop\RetailInventory\RetailInventory.csproj'.
PS C:\Users\suman\OneDrive\Desktop\RetailInventory>
```

```
PS C:\Users\suman\OneDrive\Desktop\RetailInventory> dotnet add package Microsoft.EntityFrameworkCore.Design

Build succeeded in 0.5s
info : X.509 certificate chain validation will use the default trust store selected by .NET for code signing.
info : X.509 certificate chain validation will use the default trust store selected by .NET for timestamping.
info : Adding PackageReference for package 'Microsoft.EntityFrameworkCore.Design' into project 'C:\Users\suman\OneDrive\Desktop\RetailInventory\RetailInventory.csproj'.
info : CACHE https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.design/index.json
info : CACHE https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.design/page/0.0.1-alpha/3.1.3.json
info : CACHE https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.design/page/3.1.4/6.0.0-preview.7.21378.4.json
info : CACHE https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.design/page/6.0.0-rc.1.21452.10/7.0.18.json
info : CACHE https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.design/page/7.0.19/9.0.14.json
info : CACHE https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.design/page/9.0.15/11.0.0-preview.6.26359.118.json
info : Restoring packages for C:\Users\suman\OneDrive\Desktop\RetailInventory\RetailInventory.csproj...
info : CACHE https://api.nuget.org/v3/vulnerabilities/index.json
info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.07.24.05.37.22/vulnerability.base.json
info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.07.24.05.37.22/2026.07.24.05.37.22/vulnerability.update.json
error: NU1202: Package Microsoft.EntityFrameworkCore.Design 10.0.10 is not compatible with net9.0 (.NETCoreApp,Version=v9.0). Package Microsoft.EntityFrameworkCore.Design 10.0.10 supports: net10.0 (.NETCoreApp,Version=v10.0)
error: Package 'Microsoft.EntityFrameworkCore.Design' is incompatible with 'all' frameworks in project 'C:\Users\suman\OneDrive\Desktop\RetailInventory\RetailInventory.csproj'.
```

